Double Flash Fusion

PROTOCOL DESCRIPTION DOCUMENT

IMOPA, Nancy

2026

PREPARING THE EXPERIMENTAL SET UP (BEFORE THE SUBJECT ARRIVES)

Note: The subject must be given an information form which informs them about their rights at least 24 hours before the start of the experiment.

- Prepare external electrodes (place ring stickers)
- Start computers: acq, stim, eyetracker
- Set subject ID in the 'sub00.csv' format on the stimulation scripts: DFF_EXP/TempoSpacioTask_recording.mat
- Make sure the refresh rate is at 144hz in the monitor windows setting and in the script
- Start the eyetracker
- Pre-fill Fiche de renseignements
- Fill in a seringe with gel

PREPARING THE SUBJECT

- Ask the participant to sign the consent form
- Ask the subject if S/he needs to use the bathroom
- Let the subject be seated in the room and make sure s/he is comfortable.
- Measure the size of the head and choose the appropriate cap.
- Measure the position of Cz electrode and mark it
- Place the EEG cap so that Cz is at this point and such that electrodes are symmetric L/R & back/front.
- Place external electrodes:

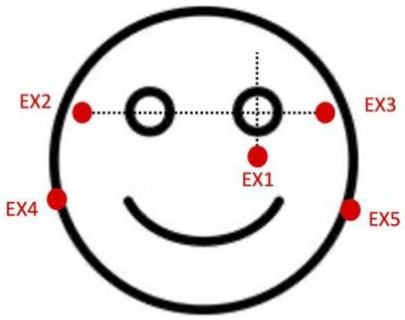


Fig 1. Place of external electrodes

- First run the DFF_EXP/TempoSpacioTask_behav.mat for then DFF_EXP/TempoSpacioTask_recording.mat

COLLECTING THE DATA

Configuring the eye-tracker

- Turn on the eye link by plugging it in
- Turn on the eyeblink host computer.
- Make sure the link is closed.
- Make sure the selected set options are as below.
- Ensure the subject's distance to the display screen is 80 cm.
- Put the marker on the subject's forehead.
- Start the camera setup and modify the parameters until you see the pupil size ok sign is green.
- Modify the lens to see that the image is no longer blurry.
- Choose both eyes.
- Choose auto threshold.
- The illuminator power must be 50 per cent.
- Turn off the lights to start experiment

EEG

- Start Biosemi ActiView and make sure the right config file is loaded (e.g. double check the sampling rate on the right panel)
- Check the impedance on Biosemi (tab « Electrode Offset ») and adjust some electrodes if needed (any > 25).
- Make sure the subject understands the task.
- In Actiview (acquisition computer) start a new recording. (keep the recording on pause)
- Naming convention: sub00.bdf

Finishing

- Feedback: Ask the subject about the experiment and note answers on the subject recording sheet.
- Offer the subjects to clean their hair
- Clean the electrodes
  (Warning: do not let the electrodes touch any metallic parts)
- Thank the subject for taking part in the experiment.

Equipment

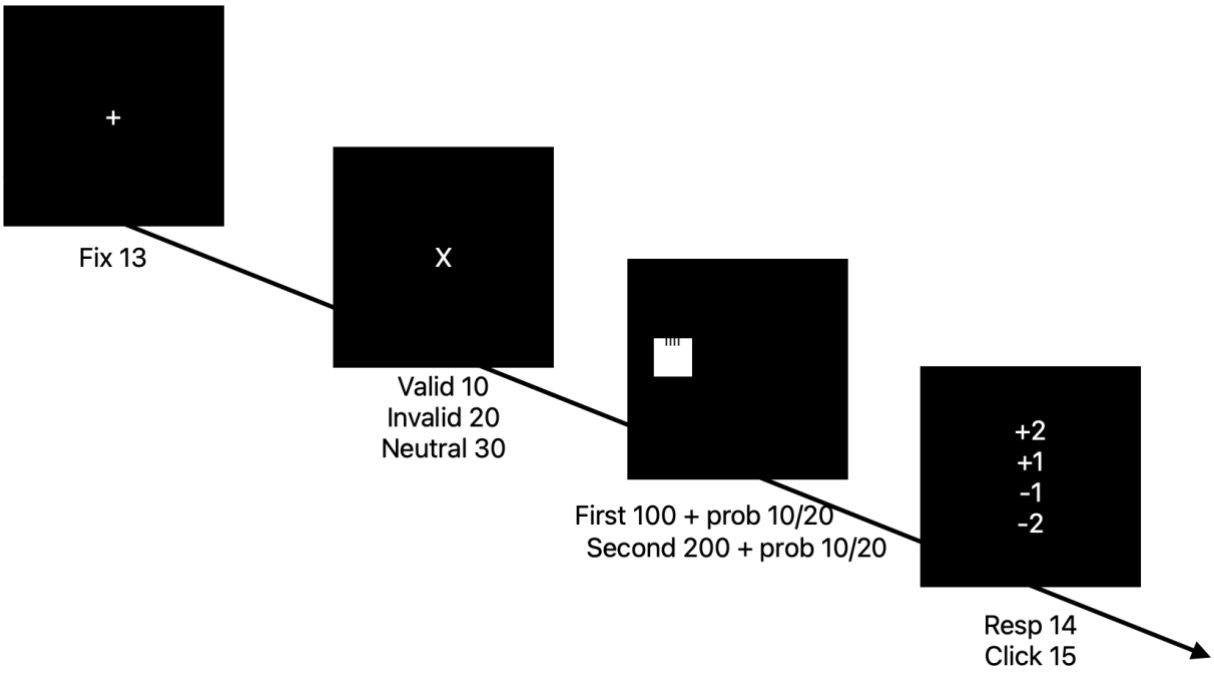
- Stimulation computer:
  - Windows: 10
  - MATLAB: 2023a
  - Screen: BENQ XL2430-B

- EEG Acquisition computer :
    - Software: ActiView902-win.vi
    - Config: Default_DFF.cfg
- Eye tracker computer :
    - Eyelink 1000 plus
    - Sampling rate: 500 Hz
- BioSemi System - 64 + 8 electrodes

Paradigm description and Triggers

Chaque essai de l'expérience comporte 4 étapes :

1. Gardez les yeux fixés sur l'écran.
2. Un indice apparaît, vous indiquant où porter votre attention tout en gardant les yeux fixés sur l'écran.
3. Un ou deux flashs apparaissent, probe lesquels se trouve une sonde, soit en haut, soit en bas.
4. Notez le nombre de flashs et de positions de probe que vous avez perçus.

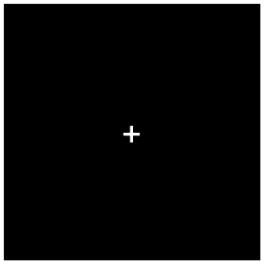


Essayez de ne pas bouger les yeux et de les garder fixés au centre, déplacez simplement votre attention.

Il y aura 380 essais. L'expérience durera environ 40 minutes. Vous pourrez faire une pause toutes les 5 minutes environ. Veuillez faire de votre mieux à chaque essai ; cela influencera les résultats.

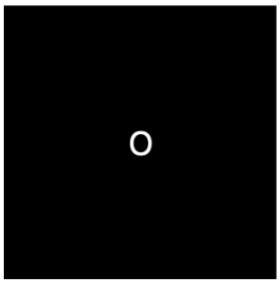
INSTRUCTION

Fixation

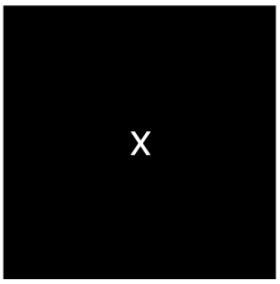


Cues

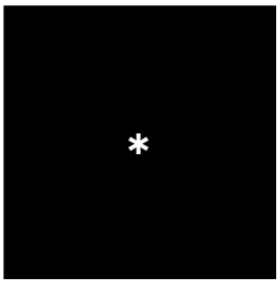
Gauche



Neutral

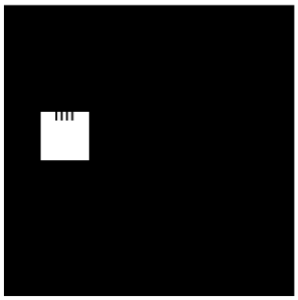


Droit

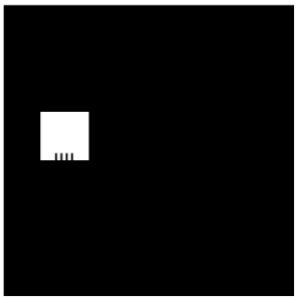


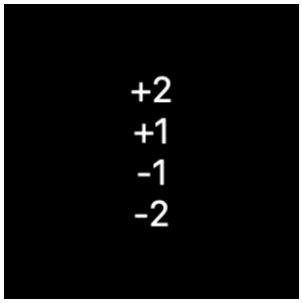
Probes

Haut



Bas



Response

- + et – indiquent la position haute et basse de la sonde.
- 1 et 2 indiquent le nombre de flashs.